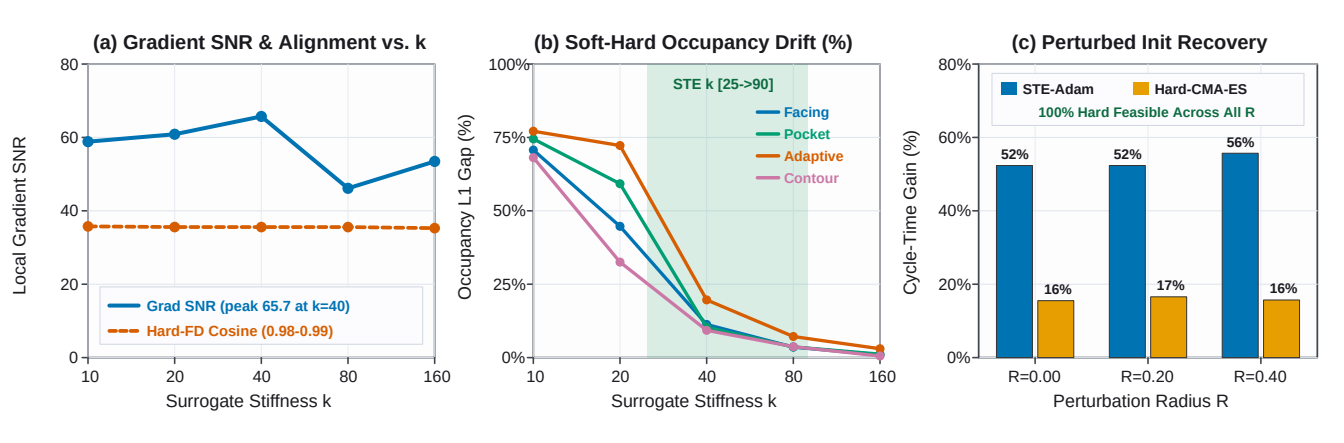
(a) Gradient SNR & Alignment vs. k
Local Gradient SNR
80
60
40
20
0
10
20
40
80
160
Surrogate Stiffness k
Grad SNR (peak 65.7 at k=40)
Hard-FD Cosine (0.98-0.99)
(b) Soft-Hard Occupancy Drift (%)
Occupancy L1 Gap (%)
100%
75%
50%
25%
0%
STE k [25->90]
Facing
Pocket
Adaptive
Contour
Surrogate Stiffness k
(c) Perturbed Init Recovery
Cycle-Time Gain (%)
80%
60%
40%
20%
0%
STE-Adam
Hard-CMA-ES
100% Hard Feasible Across All R
52%
16%
52%
17%
56%
16%
R=0.00
R=0.20
R=0.40
Perturbation Radius R